Qapture: quantum-powered carbon capture

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1 Introduction

Incorporating point-source carbon capture has the potential to significantly reduce the carbon footprint of industrial processes. We estimate that deploying this method in half of the industrial factories could lead to a reduction of up to 1.5Gt of emissions annually.

Amine-based solvents are compounds prominently used for the chemical absortion of CO_2. These solvents contain amine functional groups. A simplified carbon capture reaction involving an amine is:

$$CO_2 + 2NH_3 \rightleftharpoons NH_2COO^- + NH_4^+$$

Understanding the molecular energies of the ground states of the reactants and products is pivotal for predicting the capture efficiency of a solvent. However, this task becomes challenging for classical computers, specially as solvents increase in complexity. This leads to anharmonicity effects, and an exponential growth of the Hilbert space of the molecule, which make it difficult to model solvents efficiently. Quantum computers could reduce the complexity of the simulation from exponential to just polynomial, and allow us to find more optimal solvents.

To determine the ground state energies, the variational quantum eigensolver (VQE) is a promising candidate [3]. VQE can be implemented in analog quantum computers, such as neutral atom platforms. It is a hybrid algorithm, that blends both quantum and classical approaches. The quantum algorithm estimates the energy of the quantum system based on a chosen ansatz. Subsequently, a classical optimization subroutine provides improvements to the ansatz, to minimize the energy in the following iteration. This process converges towards the ground state energy, providing accurate results.

Neutral atom quantum computers [2] consist in an array of atoms trapped in optical tweezers (in 1D, 2D or 3D configuration). This platform naturally implements spin Hamiltonians, due to the strong dipole-dipole couplings between atoms excited to Rydberg states. Parametrized pulses of the lasers drive the system during the analog evolution, which can be viewed as a "global" gate on the entire system. In the following sections we will present the VQE and its implementation on a neutral atom quantum computer to simulate carbon capture reactions.

2 Variational Quantum Eigensolver

2.1 Outline

We are interested in simulating hamiltonians, that can be written in the Born-Oppenheimer approximation [1]:

$$\hat{H} = -\sum_i \frac{\Delta_i^2}{2} - \sum_{i,J} \frac{\mathcal{I}}{|\vec{r}_i - \vec{R}_I|} + \frac{1}{2}\sum_{i \neq j} \frac{1}{|\vec{r}_i - \vec{r}_j|}$$

We have to recreate the Hamiltonian out of an array of neutral atoms and laser pulses. This can be done through the "interaction" term, given by the strong dipole-dipole coupling between Rydberg states, and the "driving" term, induced by an external electromagnetic field [2].

$$\hat{H}_{\text{Ising}} = \sum_{i>j} \frac{C_6}{r_{ij}^6} \hat{n}_i \hat{n}_j \tag{1}$$

Where r_{ij} is the distance between the i^{th} and the j^{th} atom, and $\hat{n}_i$ is the projector on the Rydberg state. In other words, only if both i and j are in the Rydberg state there will be dipole-dipole interaction.

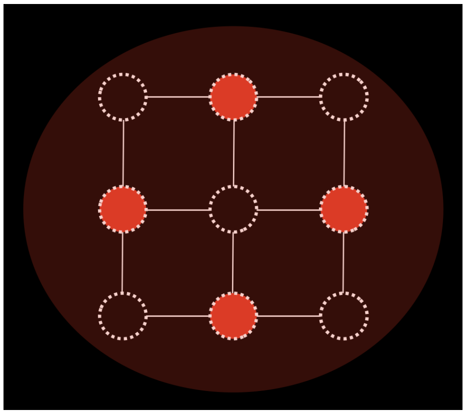

Figure 1: Ilustration of Ising interaction between atoms

Alternatively, if we choose instead as our computational basis two dipole-coupled Rydberg states, the interaction would be given by:

$$\hat{H}_{\text{XY}} = \sum_{i \neq j} \frac{C_3}{r_{ij}^3} (X_i X_j + Y_i Y_j) \tag{2}$$

Where X_i and Y_i are Pauli matrices acting on the i^{th} atom, and C_3 depends on the chosen Rydberg states. The action of this Hamiltonian corresponds to a coherent 'flip' between neighbor quantum states.

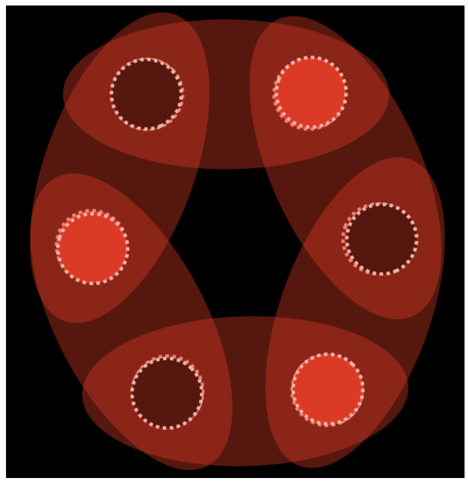

Figure 2: Ilustration of XY interaction between atoms

Note that the energy gap between the states $|0\rangle$ and $|1\rangle$ is very different depending on our basis choice; for the Ising interaction we would require visible light whereas in the 'XY' term the gap correspond to the energy of microwave radiation.

Time-dependent terms can be included in the Hamiltonian by means of a laser pulse, which corresponds to the Ising mode, and is more convenient for the neutral atom implementation. This yields the "drive" Hamiltonian:

$$\hat{H}_{\text{drive}} = \frac{\hbar}{2} \sum_{i=1}^{N} \Omega_i(t) X_i - \hbar \sum_{i=1}^{N} \delta_i(t) \hat{n}_i \tag{3}$$

Now we can define the "resource" Hamiltonian, related to our hardware, as the sum of the dipole interactions and the driving time-dependent interaction:

$$\hat{H}_R = \hat{H}_{\text{inter}} + \hat{H}_{\text{drive}} \tag{4}$$

We want to obtain the energy of the ground state of the "target" hamiltonian $\hat{H}_T$, which is time independent. Now we have the ingredients to proceed to our strategy for obtaining the ground state energy of a molecule.

The main idea is to leverage the Adiabatic theorem:
Adiabatic Theorem: A physical system remains in its instantaneous eigenstate if a given perturbation is acting on it slowly enough and if there is a gap between the eigenvalue and the rest of the Hamiltonian's spectrum.

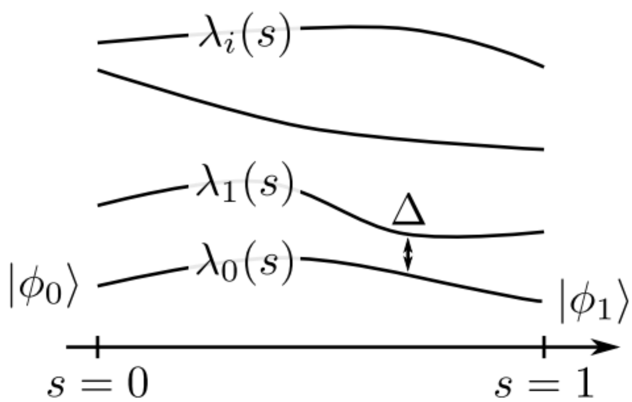

Figure 3: Evolution of eigenstates under an adiabatic trajectory

Therefore, we will proceed by preparing the ground state of a simple Hamiltonian ($\hat{H}_R$) and evolve the state according to this hamiltonian, to find the ground state of $\hat{H}_T$.

2.2 Analog implementation of VQE using Rydberg atoms

The first step will be to choose a well-known Hamiltonian whose ground state is easy to compute, and place our neutral atoms in a certain geometry which enables us to reconstruct the interactions of the real molecule using optical tweezers. Then we require digital processing to perform local qubit gates (laser beams focused in single atoms) in order to prepare the ground state $|\psi_0\rangle$.

Once we achieved $|\psi_0\rangle$, we choose an ansatz for the parameters Ω, δ, t, and then we slowly evolve the ensemble of atoms according to the "resource" Hamiltonian. After a certain time t, the resulting quantum state will be:

$$|\psi(\Omega, \delta, t)\rangle = U(t)\,|\psi_0\rangle \tag{5}$$

Where $U(t)$ is the time evolution operator given by:

$$U(t) = \mathcal{T}exp(-i\int_0^t \hat{H}_R(\Omega(\tau), \delta(\tau))d\tau) \tag{6}$$

After the unitary evolution, the result will be the ground state of the final Hamiltonian (characterized by $\Omega(t)$ and $\delta(t)$). After performing several times the experiment, we will be able to fully characterize $|\psi(\Omega, \delta, t)\rangle$. Then, we calculate its energy with respect to $\hat{H}_T$:

$$E(\Omega, \delta, t) = \langle\psi(\Omega, \delta, t)| H_T |\psi(\Omega, \delta, t)\rangle \tag{7}$$

However, that value will not be in general the ground energy of out molecule, since the final Hamiltonian (given by the chosen ansatz) does not necessarily have to be the same one as the 'target'. We perform classical optimization of Ω, δ, t and rerun the algorithm with the optimized variables, aiming to obtain a lower ground state energy in the following iteration. The optimization process can be performed via the gradient descent method in order to minimize E.

$$|\Psi_0\rangle = |101010101010\rangle$$

2.3 Simulation of CO_2 capture using VQE

We now can implement the VQE to compute the ground state energies of the reactants and products of reactions such as:

$$CO_2 + 2NH_3 \rightleftharpoons NH_2COO^- + NH_4^+$$

An important issue we must keep in mind is the number of qubits required to simulate each molecule. Depending on the ansatz chosen for the wavefunction, we will require more or less qubits to perform the simulation. According to [2], a simulation of CH_4 with the STO-3G (Slater type orbital) as basis set, requires 18 qubits and 6892 Pauli terms. We can assume similar order of qubits for CO_2 and NH_3, which makes it possible to simulate carbon capture in Pasqal's 100 qubit computer. For more complex molecules, we can keep qubit number under 100, at the expense of overly simplifying the wavefunction and reducing precision.

To choose the ansatz for the VQE we follow the alternating pulses approach described in [2], which is more scalable to bigger molecules. Then, we implement the VQE as described in Section 2.

References

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[2] Michel, A., Grijalva, S., Henriet, L., Domain, C. and Browaeys, A. [2023], 'Blueprint for a digital-analog variational quantum eigensolver using rydberg atom arrays', *Phys. Rev. A* **107**, 042602.

[3] Nguyen, M. T., Lee, Y.-L., Alfonso, D., Shao, Q. and Duan, Y. [2023], 'Description of reaction and vibrational energetics of CO2–NH3 interaction using quantum computing algorithms', *AVS Quantum Science* **5**(1), 013801.